

Adult Census Income Dataset

Machine Learning Classification Assignment

Rudhra Sitholey | 23BCY10296

VIT Bhopal University | B.Tech CSE

Dataset: UCI Adult / Kaggle | 48,842 Samples | 14 Features

Task 1

10

Dataset Understanding
Marks

Task 2

20

Data Cleaning Marks

Task 3

15

Feature Engineering
Marks

Task 4

30

Model Building Marks

Task 5

15

Performance Evaluation
Marks

This report presents a complete end-to-end machine learning pipeline on the Adult Census Income dataset. The objective is to predict whether an individual earns more than \$50,000 per year using demographic and employment features. Five classification algorithms are trained and compared across five key performance metrics.

Task 1: Dataset Understanding

[10 Marks Marks]

The Adult Census Income dataset is sourced from the 1994 US Census database and is widely used as a benchmark for binary classification. It contains 48,842 rows and 15 columns (14 features + 1 target variable). The prediction task is to determine whether a person earns more or less than \$50,000 annually.

1.1 Dataset Overview

Property	Value
Total Rows	48,842
Total Columns	15 (14 features + 1 target)
Numerical Features	6
Categorical Features	8
Target Variable	income (<=50K / >50K)
Class: <=50K	37,155 (76.07%)
Class: >50K	11,687 (23.93%)
Missing Values	Yes — encoded as "?" in raw data
Duplicate Rows	52

1.2 Feature Description

Feature	Type	Description
age	Numerical	Age of the individual (17–90)
workclass	Categorical	Employment type (Private, Self-emp, Gov, etc.)
fnlwgt	Numerical	Final weight — census sampling weight
education	Categorical	Highest education level attained
education_num	Numerical	Education encoded as number (1–16)
marital_status	Categorical	Marital status (7 categories)
occupation	Categorical	Job role (14 categories)
relationship	Categorical	Family relationship role
race	Categorical	Race of individual
sex	Categorical	Gender (Male / Female)
capital_gain	Numerical	Capital gains from investments
capital_loss	Numerical	Capital losses from investments
hours_per_week	Numerical	Weekly working hours
native_country	Categorical	Country of origin
income	Target	Binary: <=50K or >50K (prediction target)

1.3 Key Observations

- Significant class imbalance: 76% earn <=50K vs 24% earn >50K (approximately 3:1 ratio).
- capital_gain and capital_loss are highly right-skewed — most values are 0.
- education_num is a numeric encoding of the education column and carries the same information.
- The age distribution is roughly bell-shaped, peaking around 35–45 years.

- Hours per week is concentrated at 40 (standard full-time), with some extreme values.

Task 2: Data Cleaning

[20 Marks Marks]

Raw data quality issues were identified and resolved systematically before any modelling. Three categories of issues were addressed: missing values, duplicate rows, and outliers.

2.1 Missing Value Treatment

In the raw CSV, missing values are represented as "?" strings rather than NaN. These were first replaced with proper NaN values, then imputed column-by-column.

Column	Missing Count	% of Total	Treatment
workclass	2,799	5.73%	Filled with mode (Private)
occupation	2,809	5.75%	Filled with mode (Prof-specialty)
native_country	857	1.75%	Filled with mode (United-States)

Mode imputation was chosen over deletion to preserve as many samples as possible without introducing distributional bias.

2.2 Duplicate Row Removal

52 completely duplicate rows were detected using pandas drop_duplicates() and removed. The cleaned dataset contains 48,790 rows — a negligible 0.1% reduction.

2.3 Outlier Analysis (IQR Method)

Outliers were identified using the $1.5 \times \text{IQR}$ rule. They were retained rather than removed because they represent genuine real-world variation (e.g., very high capital gains, people working 80+ hours/week). Removing them would bias the model against such cases.

Feature	Outlier Count	Decision	Reason
age	215	Retained	Ages 80-90 are valid
fnlwgt	1,453	Retained	Sampling weights vary widely
education_num	1,787	Retained	Reflects genuine rare levels
capital_gain	4,035	Retained	High earners key signal
capital_loss	2,282	Retained	Valid financial transactions
hours_per_week	13,486	Retained	Over/under 40hr common

2.4 Cleaning Summary

Step	Before	After
Total rows	48,842	48,790
Missing (workclass)	2,799	0
Missing (occupation)	2,809	0
Missing (native_country)	857	0
Duplicate rows	52	0

Task 3: Feature Engineering

[15 Marks Marks]

Raw features were transformed and new features created to improve model performance. This stage includes encoding categorical variables, constructing domain-informed features, scaling, and preparing the train/test split.

3.1 Label Encoding of Categorical Variables

Eight categorical columns were transformed using LabelEncoder, converting string categories into integer codes suitable for all five classifiers.

Column	Unique Values	Encoded As
workclass	9	workclass_enc
education	16	education_enc
marital_status	7	marital_status_enc
occupation	15	occupation_enc
relationship	6	relationship_enc
race	5	race_enc
sex	2	sex_enc
native_country	42	native_country_enc

3.2 New Engineered Features

Four additional domain-informed features were created to capture patterns not directly represented in the original columns:

Feature	Formula / Logic	Rationale
capital_net	capital_gain - capital_loss	Net financial activity; single signal for investment income
age_group	Bins: 0-25, 25-35, 35-50, 50-65, 65+	Non-linear age effect on income captured as ordinal
is_married	1 if marital_status contains Married	Married individuals tend to have higher household income
high_education	1 if education_num >= 13	Bachelors degree or higher as binary flag

3.3 Feature Scaling & Train/Test Split

StandardScaler was applied to all 18 final features (zero mean, unit variance). This is critical for distance-based models (KNN, SVM) and helps Logistic Regression converge faster. Tree-based models are scale-invariant but scaling does no harm.

Parameter	Value
Total Features	18 (6 original numerical + 8 encoded categorical + 4 engineered)
Scaler	StandardScaler (zero mean, unit variance)
Test Size	20% (stratified by target class)
Random State	42 (reproducible splits)
Training Samples	39,032
Test Samples	9,758

Task 4: Model Building

[30 Marks —
Classification
Algorithms Marks]

Five classification algorithms were trained on the same preprocessed dataset. Each model was evaluated on the held-out test set (9,758 samples).

Logistic Regression

A linear probabilistic classifier that models the log-odds of the target as a linear combination of features. It serves as a strong interpretable baseline.

- `max_iter = 1000` (ensure convergence)
- `solver = lbfgs` (default, efficient for multi-class)
- `random_state = 42`

Decision Tree

A non-parametric model that partitions the feature space into rectangular regions via recursive binary splitting. Prone to overfitting; controlled here with `max_depth`.

- `max_depth = 10` (prevent overfitting)
- `criterion = gini` (default)
- `random_state = 42`

Random Forest

An ensemble of 100 decision trees trained on bootstrap samples with random feature subsets. Reduces variance via bagging. Best overall performer in this study.

- `n_estimators = 100`
- `n_jobs = -1` (parallel training)
- `random_state = 42`

K-Nearest Neighbours (KNN)

A non-parametric instance-based method that classifies a sample by majority vote of its `k` nearest neighbours. Sensitive to feature scale — hence `StandardScaler` is essential.

- `n_neighbors = 5`
- `metric = minkowski (p=2, Euclidean)`
- `weights = uniform`

Support Vector Machine (SVM)

Finds the maximum-margin hyperplane in the transformed feature space (via RBF kernel). Probability estimates enabled via Platt scaling for ROC-AUC computation.

- `kernel = rbf`
- `probability = True` (for `predict_proba`)
- `random_state = 42`

Task 5: Performance Evaluation

[15 Marks Marks]

All five models were evaluated on the same 20% test set. Five metrics are reported: Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

5.1 Metrics Explained

Metric	Formula	What it Measures
Accuracy	Correct / Total	Overall correct predictions
Precision	$TP / (TP + FP)$	Of predicted >50K, how many are correct
Recall	$TP / (TP + FN)$	Of actual >50K, how many are caught
F1 Score	$2 \times (P \times R) / (P + R)$	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve	Overall discrimination ability (0.5-1.0)

5.2 Results Table

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8457	0.7298	0.5642	0.6364	0.8996
Decision Tree	0.8592	0.7733	0.5826	0.6646	0.8966
Random Forest ★	0.8623	0.7458	0.6443	0.6913	0.9130
KNN	0.8299	0.6594	0.5985	0.6275	0.8579
SVM	0.8544	0.7680	0.5612	0.6485	0.8878

★ Best performing model across F1 Score and ROC-AUC

5.3 Analysis & Discussion

Random Forest achieves the best performance across the most important metrics — highest Accuracy (86.23%), highest F1 Score (0.6913), and highest ROC-AUC (0.9130). The ensemble of 100 trees effectively reduces variance through bagging and random feature selection, making it robust to outliers and non-linear feature interactions.

Decision Tree is the second-best model by F1 (0.6646) and achieves the highest Precision (0.7733), meaning when it predicts >50K it is most often correct — but it misses more actual >50K cases than Random Forest (lower Recall).

Logistic Regression performs surprisingly well given its simplicity, achieving a ROC-AUC of 0.8996 — nearly matching Random Forest. This suggests the income prediction problem has a strong linear component in the feature space.

SVM achieves good precision (0.7680) but lower recall (0.5612), indicating it is conservative in predicting >50K — similar to Decision Tree's trade-off pattern.

KNN performs worst overall (F1 = 0.6275, AUC = 0.8579), which is expected with 18-dimensional data. Nearest-neighbour methods suffer from the curse of dimensionality, and its performance could potentially improve with dimensionality reduction (e.g., PCA).

5.4 Key Insights

- Class imbalance (76:24) means Accuracy alone is misleading — F1 and AUC are more reliable metrics.

- capital_net, age, hours_per_week, and education_num are the top predictors (Random Forest feature importances).
- Males in this dataset earn >50K at nearly double the rate of females — a significant demographic pattern.
- Married individuals are ~3x more likely to earn >50K than unmarried individuals.
- All models achieve >82% accuracy, confirming the dataset has strong, learnable signal.

5.5 Conclusion

The Random Forest classifier is recommended as the final model for the Adult Census Income prediction task. It provides the best balance of Precision and Recall ($F1 = 0.6913$) and the strongest overall discrimination ability ($ROC-AUC = 0.9130$). Future work could explore SMOTE oversampling to handle class imbalance, hyperparameter tuning via GridSearchCV, and XGBoost as an additional benchmark.